

1. Note: these are the same matrices from 5.1 exercise 1.

a. Using the first row ($i = 1$). So the signs are $+, -, +$

$$\det \begin{bmatrix} -1 & 3 & 5 \\ 6 & 4 & 2 \\ -2 & 5 & 1 \end{bmatrix} = (-1) \det \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix} - (3) \det \begin{bmatrix} 6 & 2 \\ -2 & 1 \end{bmatrix} + 5 \det \begin{bmatrix} 6 & 4 \\ -2 & 5 \end{bmatrix} = (-1)(4 - 10) - 3(6 + 4) + 5(30 + 8) = 166$$

b. Above and in many examples we used the first row, but you can use any row (or column) you like. Since there are a row of zeros on the fourth row ($i = 4$), we'll use that row. Giving the signs $-, +, -, +$

$$\begin{aligned} \det \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & -2 & 2 & 3 \\ 0 & 0 & 6 & 2 \end{bmatrix} &= -(0) \det \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & 1 \\ -2 & 2 & 3 \end{bmatrix} + (0) \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 3 \end{bmatrix} - (6) \det \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix} + (2) \det \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix} \\ &= 0 + 0 - 6(2) + 2(4) \\ &= -4 \end{aligned}$$

2. a. $\det A = -2 \det \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} + 3 \det \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - 0 = 3$ and $\det B_2 = \det \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} = -2 \det \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} + 2 \det \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - 0 = -4$. So by Cramer's rule, $x_2 = \frac{\det B_2}{\det A} = -\frac{4}{3}$.

b. By proposition 2.4, $A^{-1} = \frac{1}{\det A} C^T$, where C^T is the *transpose* of the $C = [C_{ij}]$, the matrix

composed of A 's cofactors. The cofactor matrix is, $C = \begin{bmatrix} 6 & -4 & 5 \\ 0 & 1 & -2 \\ -3 & 2 & -1 \end{bmatrix}$. The transpose is

$$C^T = \begin{bmatrix} 6 & 0 & -3 \\ -4 & 1 & 2 \\ 5 & -2 & -1 \end{bmatrix}. \text{ So then } A^{-1} = \frac{1}{3} \begin{bmatrix} 6 & 0 & -3 \\ -4 & 1 & 2 \\ 5 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -4/3 & 1/3 & 2/3 \\ 5/3 & -2/3 & -1/3 \end{bmatrix}$$

5. a. $\det \left(\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix} - \begin{bmatrix} t & 0 \\ 0 & t \end{bmatrix} \right) = \det \begin{bmatrix} 1-t & 5 \\ 2 & 4-t \end{bmatrix} = (1-t)(4-t) - (5)(2) = t^2 - 5t - 6$

(e.) $\det \begin{bmatrix} 1-t & 1 \\ -1 & 3-t \end{bmatrix} = (1-t)(3-t) - (-1)(1) = t^2 - 4t + 4$

(f.)

$$\begin{aligned} \det \begin{bmatrix} -1-t & 1 & 2 \\ 1 & 2-t & 1 \\ 2 & 1 & -1-t \end{bmatrix} &= (-1-t) \det \begin{bmatrix} 2-t & 1 \\ 1 & -1-t \end{bmatrix} - (1) \det \begin{bmatrix} 1 & 1 \\ 2 & -1-t \end{bmatrix} + (2) \det \begin{bmatrix} 1 & 2-t \\ 2 & 1 \end{bmatrix} \\ &= (-1-t)(t^2 - t - 3) - (-t - 3) + 2(2t - 3) \\ &= -t^3 + 9t \end{aligned}$$

7. a. By assumption, $\det A = \pm 1$; so the cofactors of A are $C_{ij} = (-1)^{i+j} \det(M_{ij})$, which are all integers by exercise 6 (proved in class). Thus $C = [C_{ij}]$ has all integer entries which implies that C^T has all integer entries. A is nonsingular (since $\det A \neq 0$), so by proposition 2.4 $A^{-1} = \frac{1}{\det A} C^T = \pm C^T$. Thus A^{-1} also has all integer entries.